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PAST PAPER (2022)

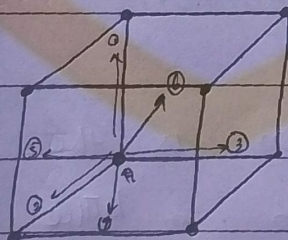
Question No: 1

(a) Coordination Number:

“The number of the nearest atoms that an atom has in unit cell is called coordination number.”

Coordination Number of SC: The simple cubic unit cell has **6** coordination number.

This can be illustrated by figure:

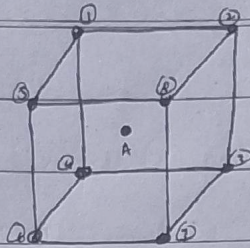


Coordination Number of BCC:

The body centered cube has coordination number **8**.

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Coordination number for FCC:-

The coordination number for FCC is **12**.

Question 1(b)

→ Symmetry Operations:

“The operation that when applied on a crystal leaves the crystal structure invariant i.e. environment of structure remain the same; is called symmetry operation.”

It has 4 types:

(i) Translation Symmetry Operation:

Translation symmetry operation is the operation which when applied to a lattice point, gives a new lattice point of identical surroundings like first lattice point.

(ii) Reflection Symmetry Operation:

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crystal is said to possess reflection symmetry about a plane if it is left unchanged in every way after being reflected by the plane.

(ii) The inversion Symmetry:

A cell have inversion symmetry if there is a point at which cell remains invariant when mathematical transformation $r \rightarrow -r$ is performed on it.

All parallel lattices are inversion symmetrical.

(iii) Rotational Symmetry:

A crystal is said to possess Rotational symmetry if crystal remains invariant after rotation through an angle θ .

$$n\theta = 2\pi$$

or

$$\theta = \frac{2\pi}{n}$$

$$; n = 1, 2, 3, 4$$
$$n \neq 5, 7.$$

(c)

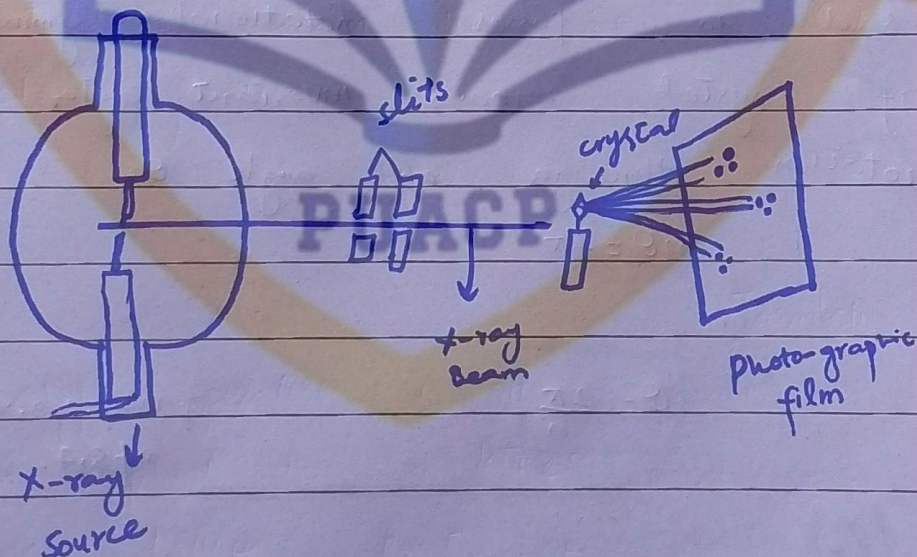
Laue Diffraction Condition:

Laue method is the oldest method of x-ray

diffraction.

A single crystal is held stationary in this method which has dimensions: $1\text{ mm} \times 1\text{ mm} \times 1\text{ mm}$.

A well collimated beam of x-rays of continuous wavelength of range $(0.2 - 2)\text{ \AA}$ is incident on specimen of a single crystal. The well collimated beam of x-rays is produced by the use of a pinhole arrangement. The reflecting planes in the crystal use a suitable wavelength of incident x-rays for which Bragg's condition $(2d \sin \theta = n\lambda)$ is satisfied. It diffracts the x-rays in different directions.



The diffraction pattern is observed on a photographic film placed at a short distance from crystal.

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The diffraction pattern consists of series of spots. The spots show the symmetry of crystal. If the crystal has four fold symmetry, then Laue's pattern will also show 4-fold symmetry.

Here are the terms involved in

Laue diffraction method:-

X-rays: Electromagnetic radiations with wavelength shorter than those of UV rays and longer than those of gamma rays.

They are commonly used in diffraction studies because they have the ability to interact with electron clouds of atoms.

Diffraction Pattern: The pattern of bright and dark spots produced when waves encounter an obstacle or aperture and undergoes diffraction. In X-rays diffraction, the diffraction pattern gives information about the arrangement of atoms in crystals.

(d)

$$\frac{2\pi}{|\mathbf{G}|}$$

Proof:- The interplanar space d_{hkl} in

a real crystal is equal to $\frac{2\pi}{|\vec{G}|}$.

Let us suppose " n " is a unit vector normal to plane (hkl) or parallel to $\vec{G}$.

$$|\vec{G}| \cdot n = (h\vec{a}^* + k\vec{b}^* + l\vec{c}^*) \cdot |n| \cos 0$$

$$= (h\vec{a}^* + k\vec{b}^* + l\vec{c}^*)$$

$$n = \frac{(h\vec{a}^* + k\vec{b}^* + l\vec{c}^*)}{|\vec{G}|}$$

$$d_{hkl} = \frac{a}{h} \cos(0) = \frac{a}{h} \cdot n$$

$$= \frac{a}{h} \cdot \frac{(h\vec{a}^* + k\vec{b}^* + l\vec{c}^*)}{|\vec{G}|} \quad \therefore \cos 0 \cdot |n| = \vec{n}$$

$$= \frac{a\vec{a}^* + 0 + 0}{|\vec{G}|} = \frac{2\pi}{|\vec{G}|} \quad \text{proved}$$

(e)

Phonon dispersion Relation for mono-atomic linear lattice.

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It is

given as:

$$\omega = 2 \sqrt{\frac{B}{m}} \sin\left(\frac{ka}{2}\right) \rightarrow (1)$$

dispersion relation gives the relation

btw the angular frequency ω and the angular wave vector k .

Velocity of wave at $k=0$:-

As,

$k=0$

$$\therefore k = \frac{2\pi}{\lambda}$$

ly:

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So, $\lambda \rightarrow \infty$.

By using equation (1) we get

Phase Velocity: The velocity of the individual wave is called phase velocity.

So,

$$V_p = \frac{\omega}{k}$$

From equ (1)

$$V_p = \frac{2}{k} \sqrt{\frac{B}{n m}} \sin\left(\frac{ka}{2}\right) \quad \because \theta \rightarrow 0$$

$$\sin \theta \approx \theta$$

$$= \frac{2}{k} \sqrt{\frac{B}{n m}} \left(\frac{ka}{2}\right)$$

$$\boxed{V_p = a \sqrt{\frac{B}{n m}}}$$

Group velocity: The velocity of group i.e. wave packet is called group velocity.

$$\therefore V_g = \frac{d\omega}{dk}$$

Diff equ (1) w.r.t k

$$\frac{d\omega}{dk} = \frac{2}{k} \sqrt{\frac{B}{n m}} \cos\left(\frac{ka}{2}\right) \cdot \frac{a}{2}$$

$$d\omega = a \sqrt{\frac{B}{n m}} \cos\left(\frac{ka}{2}\right) dk = a \sqrt{\frac{B}{n m}} dk$$

thly,

$$V_g = \frac{a \sqrt{\frac{B}{n m}} dk}{dk}$$

$$\boxed{V_g = a \sqrt{\frac{B}{n m}}}$$

At Zone Boundary:

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At Zone boundary $k = \pi/a$.

Thus, from eqn (1)

$$\omega = 2 \sqrt{\frac{B}{m}} \sin\left(\frac{\pi}{2} \cdot \frac{a}{2}\right) = 2 \sqrt{\frac{B}{m}} \sin\left(\frac{\pi}{2}\right)$$

$$\omega = 2 \sqrt{\frac{B}{m}}$$

Phase velocity: Since, $v_p = \frac{\omega}{k}$

$$v_p = \frac{2}{k} \sqrt{\frac{B}{m}} = \frac{2}{\pi/a} \sqrt{\frac{B}{m}}$$

Group velocity: $v_g = \frac{d\omega}{dk}$ Since, $v_g = \frac{d\omega}{dk}$

$$d\omega = 0$$

thus, $v_g = 0$ } (a)

From (a) and (b) it has been deduced that the group velocity at the zone boundary becomes zero.

Maximum velocity that can propagate through lattice:

For maximum velocity:-

$$\left| \sin\left(\frac{ka}{2}\right) \right|_{\max} = 1$$

So,

$$\omega_{\max} = 2 \sqrt{\frac{B}{m}}$$

In one dimensional mono-atomic lattice the maximum velocity $\omega = 2\pi f$
 $\therefore v = \lambda \omega$

that can be propagated is the speed of sound in that material.

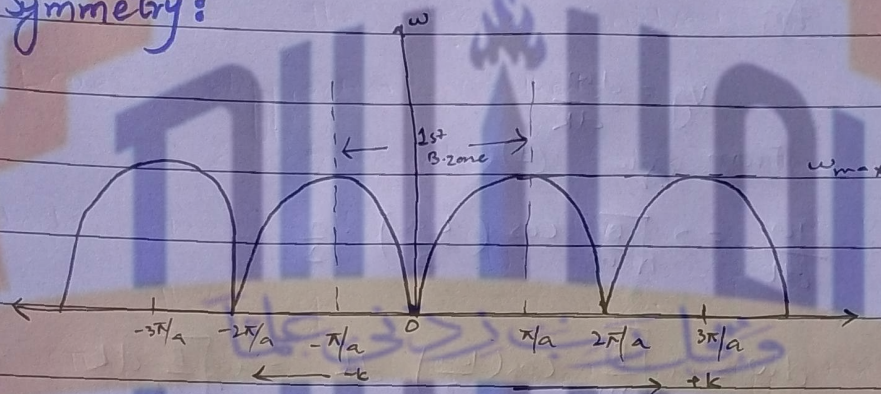
this is because in mono-atomic

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lattice; the atoms are arranged in a straight line and the propagation of sound waves occurs through the vibrations of these atoms about their equilibrium position.

Dispersion Curve with mirror symmetry:



1f)

Proof: Phase velocity = Group

velocity:

This condition is achieved when the wavelength $(\lambda \rightarrow \infty)$ and the wave vector $(k=0)$. As, the dispersion relation is:

$$\omega = 2 \sqrt{\frac{B}{m}} \sin\left(\frac{Ka}{2}\right) \quad \text{--- (1)}$$

Phase Velocity: The velocity of the individual wave is called phase velocity.

So;

$$v_p = \frac{\omega}{k} \Rightarrow \text{From eqn (1)}$$

$$v_p = \frac{2}{k} \sqrt{\frac{B}{m}} \sin\left(\frac{Ka}{2}\right) = \frac{2}{k} \sqrt{\frac{B}{m}} \left(\frac{Ka}{2}\right) \quad \text{for } \sin \theta \approx \theta$$

$$V_p = a \sqrt{\frac{B}{\rho_m}} \rightarrow (a)$$

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Group Velocity:-

The velocity of the group of waves is called group velocity.
 $\rightarrow$ (Group of Waves i.e. Wave Packet).

As,

$$V_g = \frac{d\omega}{dk}$$

From eqn (1)

$$\frac{d\omega}{dk} = 2 \sqrt{\frac{B}{\rho_m}} \cos\left(\frac{ka}{2}\right) \cdot \frac{a}{2}$$

$$d\omega = a \sqrt{\frac{B}{\rho_m}} \cos\left(\frac{ka}{2}\right) dk = a \sqrt{\frac{B}{\rho_m}} dk \quad \text{As } \cos(0) = 1$$

thus,

$$V_g = \frac{a \sqrt{\frac{B}{\rho_m}} dk}{dk}$$

$$V_g = a \sqrt{\frac{B}{\rho_m}} \rightarrow (b)$$

The equation (a) and (b) prove that for longer wavelength the medium act as non-dispersive i.e. $V_p = V_g$.

Answer the following questions.

Q2(a)

Madelung Energy :

The long range interaction between ions with charge $\pm q$ is the electrostatic interaction $\pm q^2/r$, attractive between ions of opposite charge and repulsive between ions of the same charges. The main contribution to the binding Energy of ionic crystals is electrostatic and is called Madelung Energy.

Calculations:

The sum of all interaction for the i^{th} atoms can be written as :

$$U_i = \sum'_{j \neq i} U_{ij} \quad \text{--- (1)}$$

where, the primed summation includes all ions except $j=i$

$$U_{ij} = \lambda \exp(-\delta_{ij}/\rho) \pm q^2/\delta_{ij} \quad \text{--- (2)}$$

where, $\pm$ represents interactions between like and opposite charges.

$\lambda \exp(-\delta_{ij}/\rho)$ = Central field Repulsive Potential.

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$\lambda, \rho =$ Empirical Parameters

$\Rightarrow$ It is convenient to introduce dimensionless Parameter

$\Rightarrow$ where, R is the nearest neighbor separation in the crystal.

As,

$$\delta_{ij} = \rho_{ij} R$$

$\Rightarrow$ The equation no 2 will become

$$U_{ij} = \lambda \exp\left(-\frac{\rho_{ij} R}{\rho}\right) + \frac{Cv^2}{\rho_{ij} R}$$

$\Rightarrow \rho_{ij} = 1$ for Repulsive interaction. As it is very short range for first neighbor ion.

So,

$$U_{ij} = \lambda \exp\left(-\frac{R}{\rho}\right) + \frac{Cv^2}{\rho_{ij} R} \quad \text{--- (3)}$$

$\Rightarrow$ The total Lattice energy U_{tot} of a crystal is composed of N molecules.

$$U_{\text{tot}} = NU_i$$



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So, equation no. (3) become

$$U_{tot} = NU_i = NZ\lambda \exp(-R/\rho) \pm \sum_j \frac{q_j^2}{P_{ij}R} \dots (4)$$

⇒ Where Z is the number of nearest neighbours of any ion.

Madelung Constant:

⇒ From equation no. (4)

$$\alpha = \sum_j \pm \frac{1}{P_{ij}}$$

Where the sign + is taken for the like charges and sign - for the unlike charges.

The value of the Madelung constant is of Central importance in the theory of ionic crystals.

⇒ Equation no. (4) become

$$U_{tot} = N \left[Z\lambda \exp(-R/\rho) - \alpha \frac{q^2}{R} \right] \dots (5)$$

At Equilibrium Separation:

$$\left(\frac{dU_{\text{tot}}}{dR} \right)_{R=R_0} = 0$$

$$\frac{dU_{\text{tot}}}{dR} \Rightarrow N \frac{d}{dR} \left[Z\lambda e^{-R/\rho} - \alpha q v^2 / R \right] = 0$$

$$N \left[\frac{d}{dR} \left(Z\lambda e^{-R/\rho} \right) \cdot (-1/\rho) + \alpha \frac{d}{dR} \left(\frac{q v^2}{R} \right) \right] = 0$$

$$-\frac{Z\lambda e^{-R/\rho}}{\rho} + \alpha \frac{q v^2}{R^2} = 0$$

$$\alpha q v^2 / R^2 = \frac{Z\lambda e^{-R/\rho}}{\rho}$$

$$\lambda Z e^{-R/\rho} = \alpha q v^2 \rho / R^2 \quad \text{--- (6)}$$

$$R_0^2 e^{-R_0/\rho} = \rho \alpha q v^2 / Z\lambda$$

$\Rightarrow$ The equilibrium separation R_0 can be determined if ρ , λ of Repulsive interaction are known.

Put equation no. (6) in equation (5)

$$U_{\text{tot}} = N \left(\alpha q v^2 \rho / R_0^2 \right) - \alpha q v^2 / R_0$$

$$U_{\text{tot}} = -N\alpha q^2/R_0 \left(1 - \rho/R_0\right)$$

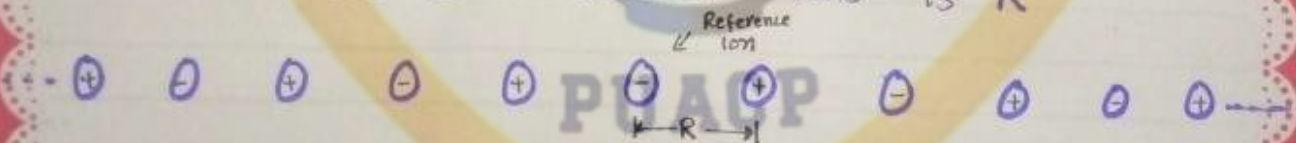
The term $-N\alpha q^2/R_0$ is the

Madelung Energy. The ρ can be determined of the order has a very short Repulsive interaction range.

Evaluation of Madelung Constant for an infinite one dimensional array of ions:

$$\alpha = \sum_j \pm \frac{1}{r_{ij}}$$

$\Rightarrow$ A line of ions of alternating signs with distance between ions is R



By dividing R on both sides of equation (1)

$$\frac{\alpha}{R} = \sum_j \pm \frac{1}{r_{ij}R}$$

$$\therefore r_{ij} = R$$

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$$\frac{\alpha}{R} = \sum_j \frac{\pm}{0.ij}$$

$$\frac{\alpha}{R} = \left[\frac{1}{R} - \frac{1}{2R} + \frac{1}{3R} - \frac{1}{4R} + \dots + \frac{1}{R} - \frac{1}{2R} + \frac{1}{3R} - \frac{1}{4R} + \dots \right]$$

$$\frac{\alpha}{R} = \left[\frac{2}{R} - \frac{2}{2R} + \frac{2}{3R} - \frac{2}{4R} + \dots \right]$$

$$\frac{\alpha}{R} = \frac{2}{R} \left[1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \right]$$

$$\frac{\alpha}{R} = \frac{2}{R} \ln(1+1)$$

$$\frac{\alpha}{R} = \frac{2}{R} \ln 2$$

$$\alpha = 2 \ln 2$$

$$\alpha = 2 \ln 2$$

$$\alpha = 1.386$$

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Values of Madelung Constant are:

$$\text{NaCl} \rightarrow \alpha = 1.7475$$

$$\text{CsCl} \rightarrow \alpha = 1.7626$$

$$\text{ZnS} \rightarrow \alpha = 1.6381$$

Q2(b)

Cohesive Energy

The cohesive energy of a solid is defined as;

"The energy required to separate constituent atoms apart from each other and to bring them to an assembly of neutral free atoms."

Mathematically:-

Total energy (attractive and repulsive) of two atoms of inert gas is give by

$$U = U_{\text{attractive}} + U_{\text{repulsive}}$$

$$U = -\frac{A}{R^6} + \frac{B}{R^{12}}$$

$\Rightarrow$ Let us introduce two new parameters ϵ and σ .

$\Rightarrow$ Such that

$$A = 4\epsilon\sigma^6$$

$$B = 4\epsilon\sigma^{12}$$

then;

$$U = -\frac{4\epsilon\sigma^6}{R^6} + \frac{4\epsilon\sigma^{12}}{R^{12}}$$

$$U = 4\epsilon \left[\left(\frac{\sigma}{R} \right)^{12} - \left(\frac{\sigma}{R} \right)^6 \right] \quad \text{---(1)}$$

This is known as Lennard-Jones

interaction.

System of N-Atoms:-

Let U_{ij} be the Potential energy of interaction between the atoms "i" and "j" of the crystal. The total binding energy of interaction U_i of the atoms i due to all other atoms of the crystal will be

$$U_i = \sum_j U_{ij}$$

⇒ Where Summation extends over all the atoms excluding $j=i$.
 U_{ij} consist of a following two terms.

- (a) Repulsive energy which is given by $\frac{B}{(R_{ij})^{12}}$, where R_{ij} is the distance between ith and jth atom.

(b) Attractive energy which is given by $\frac{A}{(R_{ij})^6}$.

$\Rightarrow$ The total binding energy of interaction U_{ij} between atoms i & j of the crystal is then

$$U_{ij} = \frac{B}{R_{ij}^{12}} - \frac{A}{R_{ij}^6}$$

Which can be written by

$$U_{ij} = 4e \left[\left(\frac{\sigma}{R_{ij}} \right)^{12} - \left(\frac{\sigma}{R_{ij}} \right)^6 \right]$$

$\Rightarrow$ If the crystal is composed of N -atoms, then the total binding-energy of the crystal will be;

$$U = \frac{1}{2} N U_{ij}$$

$$U = \frac{1}{2} N(4E) \left[\sum_j \left(\frac{\sigma}{R_{ij}} \right)^{12} - \left(\frac{\sigma}{R_{ij}} \right)^6 \right] \quad (2)$$

⇒ The factor $\frac{1}{2}$ used because the interaction between the atoms i and j has to be counted only one. Let us introduce a dimensionless quantity P_{ij} such as;

⇒ $R_{ij} = P_{ij} R$ where " R " is the interatomic distance.

Eq (2) can be written as:-

$$U = 2NE \left[\sum_j \left(\frac{\sigma}{R_{ij}} \right)^{12} - \sum_j \left(\frac{\sigma}{R_{ij}} \right)^6 \right]$$

In case of $R_{ij} = P_{ij} R$

$$U = 2NE \left[\sum_j \left(\frac{\sigma}{P_{ij} R} \right)^{12} - \left(\frac{\sigma}{P_{ij} R} \right)^6 \right]$$

⇒ In case of FCC Structure;

$$\sum_j \left(\frac{1}{r_{ij}} \right)^{12} = 12.13188$$

$$\sum_j \left(\frac{1}{r_{ij}} \right)^6 = 14.45392$$

Hence:-

$$U = 2NE \left[(12.13188) \left(\frac{\sigma}{R} \right)^{12} - (14.454) \left(\frac{\sigma}{R} \right)^6 \right] \quad (3)$$

• Equilibrium Lattice Constant

If we neglect the kinetic energy of the inert gas crystals atom the cohesive energy of an inert gas crystal is given by summing the Lennard-Jones Potential over all pair of atoms

in the crystal the total potential energy is;

$$U_{\text{total}} = \frac{1}{2} N 4\epsilon \left[\sum_j \left(\frac{\sigma}{P_{ij}R} \right)^{12} - \sum_j \left(\frac{\sigma}{P_{ij}R} \right)^6 \right]$$

⇒ Where P_{ij} is the distance between reference atom "i" and any other atom "j" expressed in terms of the nearest-neighbor distance "R". The factor $1/2$ occurs with the N to compensate twice each pair of atom.
For FCC structure,

$$\sum_j P_{ij}^{-12} = 12.13188$$

$$\sum_j P_{ij}^{-6} = 14.45392$$

⇒ There are 12 nearest-neighbor sites in the fcc structure. The nearest neighbor contribute most of the interaction energy of inert gas crystals. If we take U_{tot} as

as the total energy of the crystal.
At equilibrium interatomic distance at; $R = R_0$ the P.E is minimum.

Therefore we are to find $\frac{dU}{dR}$

at $R = R_0$

$$\left(\frac{dU}{dR}\right)_{R=R_0} = 2NE \left\{ (12)(12 \cdot 13188)(-1) \frac{\sigma^{12}}{R_0^{13}} - (6)(14 \cdot 454)(-1) \frac{\sigma^6}{R_0^7} \right\}$$

$\Rightarrow$ Since energy is minimum at $R = R_0$ So,

$$\left(\frac{dU}{dR}\right)_{R=R_0} = 0$$

$$2NE \left\{ 6(14 \cdot 454) \frac{\sigma^6}{R_0^7} - 12(12 \cdot 13188) \frac{\sigma^{12}}{R_0^{13}} \right\} = 0$$

$$6(14 \cdot 454) \frac{\sigma^6}{R_0^7} = 12(12 \cdot 13188) \frac{\sigma^{12}}{R_0^{13}}$$

$$\left(\frac{\sigma}{R_0}\right)^6 = \frac{6(14.454)}{12(12.13188)}$$

$$\left(\frac{R_0}{\sigma}\right)^6 = \frac{2(12.13188)}{14.454}$$

$$\left(\frac{R_0}{\sigma}\right)^6 = 1.6786$$

$$\frac{R_0}{\sigma} = (1.6786)^{1/6}$$

$$\frac{R_0}{\sigma} = 1.09 \Rightarrow \text{Put in eq (3)}$$

$$U_0 = 2NE \left[(12.13188)(1.09)^{-12} - (14.454)(1.09)^6 \right]$$

After Solving ;

$$U_{\text{tot}}(R_0) = (-2.15)(4NE)$$

$$U_{\text{tot}} R_0 = (-2.15)(4NE)$$

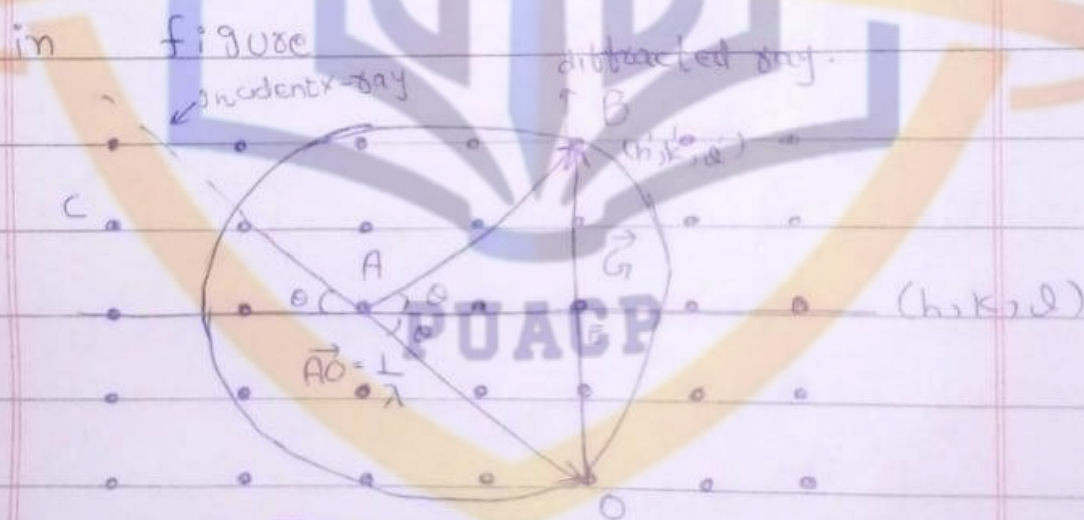
This value is same for all inert gases. This is the calculated cohesive energy when the atoms are at rest.

Question # 03

(a)

What is vector form of Bragg's law? Drive its equation and explain its importance / Physical Significance in Solid State of Physics.

Consider a reciprocal lattice space which has a large no of Reciprocal lattice points as shown in figure



Fig(a)

Vector Form of Bragg's Law:-

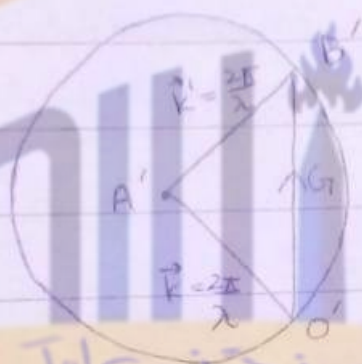
To convert $\vec{A}$ into vector $\vec{K}$, multiply $\vec{A}$ by 2π .

$$\text{As } \vec{A} = \frac{1}{\lambda}$$

Multiplying by 2π

$$\vec{AO} = 2\pi \times \frac{1}{\lambda} = \frac{2\pi}{\lambda} = \vec{k} \quad \therefore \frac{2\pi}{\lambda} = \vec{k}$$

For this $\vec{AB}$ vector also multiplied by 2π , so radius of circle will increase 2π times.



$\vec{A'O'}$ is 2π times greater than $\vec{AO}$.

with the help of Ewald construction, we can write Bragg's law in vector form. Let all vectors drawn in fig (a) are magnified by multiple of 2π .

$$\vec{A'O'} = \vec{k} = \text{Incident wave vector.}$$

$$|\vec{k}| = 2\pi/\lambda$$

$$\vec{A'B'} = \vec{k'} = \text{Diffracted wave vector.}$$

$$|\vec{k'}| = \frac{2\pi}{\lambda}$$

Using triangle law of vector addition:-

$$\vec{A'O'} + \vec{O'B'} = \vec{A'B'}$$

$$\vec{K} + \vec{G} = \vec{K'}$$

Since $\vec{A'O'} = \vec{A'B'}$ so $\vec{K} = \vec{K'}$

then

$$\vec{K} + \vec{G} = \vec{K}$$

Taking square on both side

$$(\vec{K} + \vec{G})^2 = (\vec{K})^2$$

$$(\vec{K})^2 + (\vec{G})^2 + 2\vec{K}\vec{G} = \vec{K}$$

$$(\vec{G})^2 + 2\vec{K}\vec{G} = 0$$

above eq. is known as
Bragg's law in vector form.

(b)

what are Quasi crystals?
Explain with at least two examples.

QUASI CRYSTALS:-

Quasi crystals are a type of structure that exhibits ordered patterns but lacks the **Periodicity** found in traditional crystals. They have unique symmetrical properties that are not repeatable in a regularly repeating lattice. This means they have long range order like crystals, but their atomic arrangement doesn't repeat at regular intervals.

Examples:-

- Aluminium-Manganese (Al-Mn)

Quasi crystals:-

(Al-Mn) Quasi

Crystals discovered in the **1980s** by **Dan Shechtman**, Al-Mn was first experimentally observed quasi crystals. They have an icosahedral symmetry, meaning their structure resembles a **20-sided polyhedron**.

This symmetry is impossible in traditional periodic crystals.

● **Decagonal Quasicrystals:-**

These Quasicrystals have a **ten-fold** rotational symmetry and were first discovered in **1987**. They exhibit a pattern that repeats every tenth atomic layer, but the overall structure lacks periodicity. One well known example of a decagonal quasicrystal is found in an alloy of aluminium, nickel and cobalt.

Question No 4

(a)

Define lattice heat capacity. What is Einstein model of lattice heat capacity? Drive its expression and compare it with experimental Results.

Lattice Heat Capacity:-

Lattice heat capacity refers to the specific heat capacity of the lattice structure of material. It represents the amount of heat energy required to raise the temperature of lattice by a certain amount.

$$C = dQ/dT$$

Einstein Model:-

Einstein model is a model of a solid based on two ~~assumpti~~ assumptions.

- Each atom in a lattice is an independent **3D** quantum harmonic oscillator.

- All atoms oscillate with same **frequency ω** (contrast with the Debye Model).

The allowed energy values of a 1-D quantum harmonic oscillator of angular frequency ω are discrete given by

$$E_n = \left(n + \frac{1}{2}\right) \hbar \omega \quad n = 0, 1, 2, \dots$$

Neglecting the zero point energy as this not affect the final result so,

$$E_n = n \hbar \omega \quad n = 1, 2, 3, \dots$$

The average energy of each harmonic oscillator is given by.

$$\bar{E} = \frac{\sum_{n=1}^{\infty} N_n E_n}{\sum_{n=1}^{\infty} N_n}$$

N_n = no of oscillator in n^{th} state.

According to statistical Mechanics.

$$N_n = N_0 \exp\left(-\frac{E_n}{k_B T}\right)$$

$$\bar{E} = \frac{\sum_{n=0}^{\infty} E_n N_0 \exp(-E_n/k_B T)}{\sum_{n=0}^{\infty} N_0 \exp(-E_n/k_B T)}$$

$$\bar{E} = \frac{\sum_{n=0}^{\infty} n \hbar \omega \exp\left(-\frac{n \hbar \omega}{k_B T}\right)}{\sum_{n=0}^{\infty} \exp\left(-\frac{n \hbar \omega}{k_B T}\right)} \quad \because E_n = n \hbar \omega$$

$$\bar{E} = \hbar \omega \frac{\sum_{n=0}^{\infty} n \exp(-n \hbar \omega / k_B T)}{\sum_{n=0}^{\infty} \exp(-n \hbar \omega / k_B T)} \quad (1)$$

Let $\exp(-\hbar \omega / k_B T) = x$

So $\exp\left(-\frac{n \hbar \omega}{k_B T}\right) = x^n$

So eq (1) will become

$$\bar{E} = \hbar \omega \frac{\sum_{n=0}^{\infty} n x^n}{\sum_{n=0}^{\infty} x^n} \quad (2)$$

Here

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots \text{infinite G.P series.}$$

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

$$\begin{aligned} \sum_{n=0}^{\infty} n x^n &= x \frac{d}{dx} \sum_{n=0}^{\infty} x^n \\ &= x \frac{d}{dx} \left(\frac{1}{1-x} \right) = \frac{x}{(1-x)^2} \end{aligned}$$

So eq (2) becomes.

$$\bar{E} = \frac{\hbar \omega x}{(1-x)^2} \div \frac{1}{(1-x)}$$

$$\bar{E} = \frac{\hbar \omega x}{(1-x)^2} \times \frac{(1-x)}{1} = \frac{\hbar \omega x}{1-x}$$

$$\bar{E} = \hbar \omega \frac{1}{\frac{1}{x} - x} = \hbar \omega \frac{1}{\frac{1}{x} - 1}$$

As $x = \exp\left(-\frac{\hbar \omega}{k_B T}\right) \Rightarrow \frac{1}{x} = \exp\left(\frac{\hbar \omega}{k_B T}\right)$

So

$$\bar{E} = \hbar \omega \frac{1}{\exp\left(\frac{\hbar \omega}{k_B T}\right) - 1}$$

The total energy of 1 mole of atoms of a solid (having N atoms) is

$$E = 3N \bar{E}$$

So $E = 3N \hbar \omega \frac{1}{\exp\left(\frac{\hbar \omega}{k_B T}\right) - 1}$

Thus molar specific heat at constant volume is given by

$$C_v = \left(\frac{dE}{dT}\right)_v$$

$$C_v = 3N \hbar \omega \frac{d}{dT} \left(\frac{1}{\exp\left(\frac{\hbar \omega}{k_B T}\right) - 1} \right)$$

$$C_v = 3N\hbar\omega \left[\frac{-1}{\left[\exp\left(\frac{\hbar\omega}{k_B T}\right) - 1 \right]^2} \cdot \exp\left(\frac{\hbar\omega}{k_B T}\right) \cdot \left(-\frac{\hbar\omega}{k_B T^2}\right) \right]$$

$$C_v = 3N \left[\frac{\hbar^2 \omega^2}{k_B T^2} \times \frac{\exp(\hbar\omega/k_B T)}{\left[\exp(\hbar\omega/k_B T) - 1 \right]^2} \right]$$

Multiply and divide by k_B

$$C_v = 3Nk_B \left(\frac{\hbar\omega}{k_B T} \right)^2 \frac{\exp(\hbar\omega/k_B T)}{\left[\exp(\hbar\omega/k_B T) - 1 \right]^2} \quad (3)$$

$$Nk_B = R$$

and we define

$$\theta_E = \frac{\hbar\omega}{k_B} \quad (\text{called Einstein temp})$$

By putting R and θ_E in (3) we get.

$$C_v = 3R \left(\frac{\theta_E}{T} \right)^2 \frac{e^{\theta_E/T}}{(e^{\theta_E/T} - 1)^2} \quad (4)$$

Special Cases:-

Now we consider two temperature ranges.

*** High temperature Ranges-**

At high temperature.

$$\theta_E \ll T \quad \text{hence} \quad e^{\theta_E/T} = 1 + \frac{\theta_E}{T}$$

So eq (4) becomes

$$C_v = 3R \left(\frac{\theta_E}{T} \right)^2 \frac{1 + \theta_E/T}{(\theta_E/T)^2}$$

$$C_v = 3R \left(1 + \frac{\theta_E}{T} \right) \quad \text{--- (5)}$$

Since $\theta_E \ll T \Rightarrow \frac{\theta_E}{T} \ll 1$

So $\frac{\theta_E}{T} \rightarrow 0$ putting in (5)

we get

$$C_v = 3R(1 + 0)$$

$$C_v = 3R$$

which is Dulong Petit's law obtained from classical theory.

So at high temp Einstein theory agree with classical theory.

Low Temperature Range:-

At low temp $\theta_E \ll T$

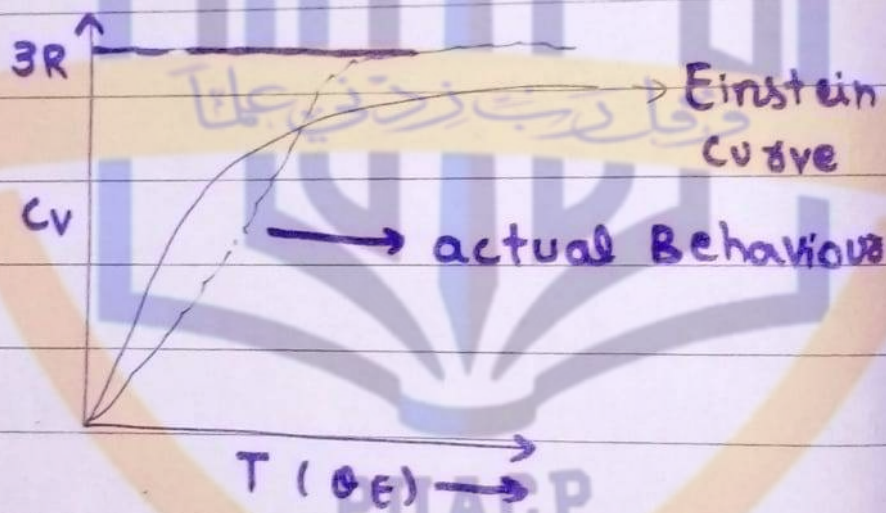
and hence $e^{\theta_E/T} - 1 \approx e^{\theta_E/T}$

So eq (4) becomes

$$C_v = 3R \left(\frac{\theta_E}{T} \right)^2 \frac{e^{\theta_E/T}}{(e^{\theta_E/T})^2}$$

$$C_v = 3R \left(\frac{\theta_E}{T} \right)^2 \cdot e^{-\theta_E/T}$$

The exponential term more important than the term $\left(\frac{\theta_E}{T} \right)^2$ in determining the variation of specific heat C_v with temperature, Thus with decreasing temperature, C decreases exponentially.



The Einstein specific heat curve is fairly close to experimentally curve except at low temperature. In this region, the Einstein specific heat approaches more rapidly than

observed values. However, at high temperature it approaches to classical value.

